

Objectives:

- Knowing the difference between Software engineering vs Programming
- Discussing various topics within Software Engineering and its importance.
- Practice critical thinking and communication skills

Task 1 / Exercise C1 (Introductory):

1. What are the major tasks in developing and supporting a software system?
2. What is the difference between a client and an end-user?
3. Discuss whether you think a programming language constraint may be viewed as a requirement. Explain why you think so.
4. Explain why developing a software product is a difficult task.

Task/ Exercise C2 (Roles & Responsibilities)

1. List the possible roles or positions within a software team.
2. Scenario: The coffee shop owner, in the Quick Cup app, wants to add a "Happy Hour" with a 25% discount on certain drinks from 2-4 PM. To which roles would you assign the tasks of:
 - Clarifying the exact rules with the owner?
 - Designing how the discount appears in the app?
 - Implementing the time-based discount logic?
 - Testing that the discount works correctly and doesn't apply on weekends?
 - Making sure the feature is delivered on schedule?

Task/ Exercise C3 : (Qualities)

A "good" software product has many quality attributes (Correctness, Reliability, Usability, Performance, Security, etc.). However, for any given project, some are more critical to success than others.

- For the "Quick Cup" Coffee App, rank the following five quality attributes from 1 (most important) to 5 (least important) for the project's success. Be prepared to defend your ranking.

Quality Attribute	Your Rank (1-5)	Justification for your ranking
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Reliability (Does the app crash? Does it ever lose an order?)		
Maintainability (How easy is it for developers to add a new "seasonal drink" feature in six months?)		
Usability (Is it easy and intuitive for students to order?)		
Security (Is the "Pay at Counter" system safe from basic hacks?)		
Performance (Is the app fast and responsive, especially during the busy 10 minutes between classes?)		

Task / Exercise C4: (Classification)

Background: As discussed in the lecture (Bertrand Meyer's ABC Classification), not all software is created equal. The engineering effort must match the criticality of the system.

- Type A (Acute): Failure can have catastrophic consequences (e.g., loss of life).
- Type B (Business): Failure has significant business or financial impact.
- Type C (Casual): Failure is an inconvenience with minimal impact.

Instructions: In your teams, classify the following systems and provide a brief justification for your choice.

System	Classification (A, B, or C)	Justification (Why?)
"Quick Cup" Coffee App		
Software for an airplane's auto-pilot system		

A personal website for a student to showcase their portfolio		
A hospital's patient information and management system		
An online banking system for a major bank		
A popular online multiplayer video game (e.g., Fortnite)		
Software controlling a city's power grid		
A mobile app that reminds you to water your plants		

Task/ Exercise 5: The Real World (Ethical/Practical Dilemma)

Instructions: Read the following scenario and discuss how your team would respond.

The owner of the coffee shop is very excited but also on a very tight budget. To save money, he makes a suggestion: *"Let's just build the student app first. We can skip the barista tablet view for now. The baristas can just read the orders from my phone as they come in. It'll be fine!"*

1. As a software engineer, what are the potential risks and problems with the client's suggestion? (Think about the "system," not just the code).
2. How would you explain these risks to the client in a simple, non-technical way? What principle from the lecture (Davis's principals) applies here?

Optional exercises to work on beyond class

Exercise P1 (Optional) :(Risks and Projects) :

Within your team, discuss and answer the following questions,

1. Why does it take so long to get software finished?
2. Why are development costs so high?
3. Why can't we find all errors before we give the software to our customers?

4. Why do we spend so much time and effort maintaining existing programs?
5. Why do we continue to have difficulty in measuring progress as software is being developed and maintained?
6. In your opinion, what are the main reasons of failure when developing a software product

Exercise P2 (Optional) :

Many organizations buy commercial software, thinking it is cheaper than developing and maintaining software in-house. Describe the pros and cons of using COTS (Commercial off-the-shelf) software. For example, what happens if the COTS products are no longer supported by their vendors? What must the customer, user, and developer anticipate when designing a product that uses COTS software in a large system?

Exercise P3 (Optional) :

What is the most important difference between generic software product development and custom software development? What might this mean in practice for users of generic software products?

Exercise P4 (Optional) :

Explain why keeping all members of a group informed about progress and technical decisions in a project can improve group cohesiveness.

Exercise P5 (Optional) :

Your manager asks you to deliver software to a schedule that you know can only be met by asking your project team to work unpaid overtime. All team members have young children. Discuss whether you should accept this demand from your manager or whether you should persuade your team to give their time to the organization rather than to their families. What factors might be significant in your decision?

Exercise P6 (Optional) :

1. You are a member of the user interface team. You are responsible for designing and implementing forms that collect information about users of the system (e.g., first name, last name, address, E-mail address, level of expertise). The information you are collecting is stored in the database and used by the reporting subsystem. You are not sure which fields are required information and which are optional. How do you find out?
2. You have been reassigned from the user interface team to the database team due to staff shortages and replanning. The implementation phase is well underway. In

which role would you be most proficient, given your knowledge of the user interface design and implementation?

Exercise P7 (Optional):

You have completed your degree at the university and been employed at a company. The company needs a full software solution to run their operations (Sales, Purchase and HR). They asked you for a decision of whether they buy an existing software solution or you develop a new software product (in-house) for the company with a team of software developers working for the same company.

What would be your answer and justify ?